**RS minimal model: floor summary and literature reproductions**
working notes, June 2026. Quark sector, minimal (non-custodial) Randall-Sundrum.

Quick notes after the constraint-code audit and fixes. Two things here: (1) the floor summary (what $\mathrm{M_{KK}}$ the minimal model actually needs), and (2) our reproductions of the relevant published figures, side by side with the paper. Everything is the minimal (non-custodial) RS model in the S1 anarchic scenario, so in each comparison the paper panel on the left is cropped to the SINGLE minimal/S1 panel we actually reproduce (the custodial panels and the other scenarios in those figures are deliberately not shown). All run on the fixed code.

**Lane note (important).** These reproductions are the **anarchic baseline (lane A)** – the literature strawman – used to validate the code against published figures. They are **NOT the production / our model**: production as run (lane B, the fit-aligned MFV without V5KM) is a sharp $\varepsilon_K$ wall at about **7 TeV**, and the FPR ideal (lane C, V5KM aligned) is about **2 TeV**. So the ∼30 TeV anarchic $\varepsilon_K$ cloud below is the strawman, not "our floor". Canonical three-lane definitions: `docs/MODEL_CONVENTIONS.md`.

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## 1. Floor summary

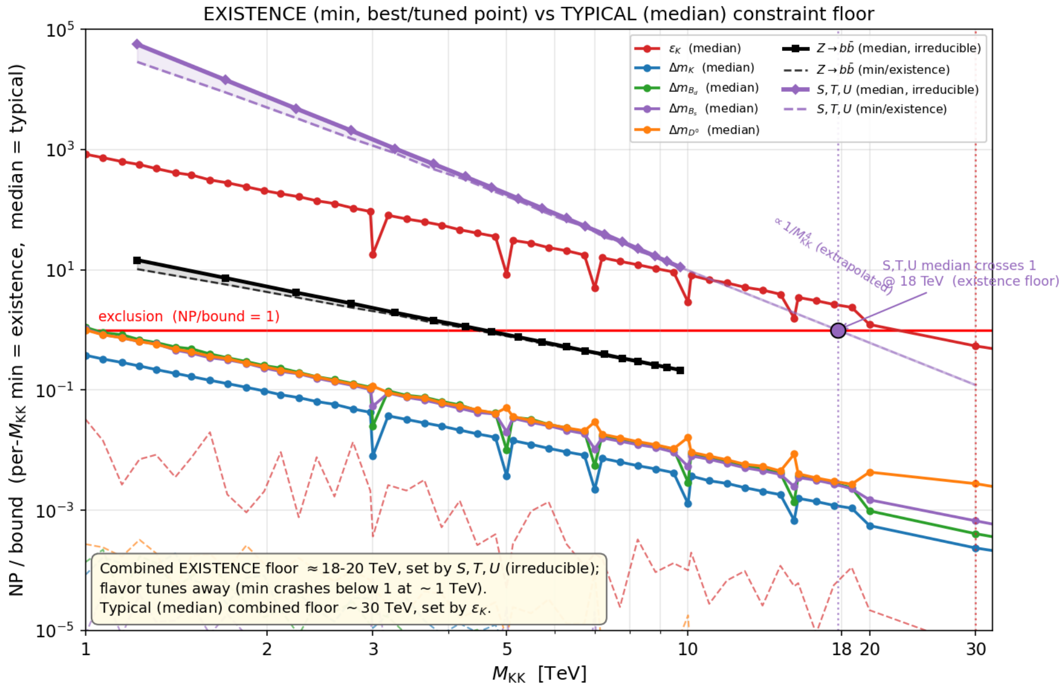
There are two honest notions of a floor, and they are very different:

- **Typical floor:** where the *median* anarchic point becomes allowed. This is the usual literature statement (**lane A, anarchic strawman – NOT our model**). With current data it is about **30 TeV**, set by $\varepsilon_K$. (Production as run, lane B, is a sharp $\varepsilon_K$ wall at ∼7 TeV; the FPR ideal, lane C, is ∼2 TeV.)
- **Existence floor:** where even a single best-case (fine-tuned) point is allowed. This is about **18 to 20 TeV**, set by $S, T, U$.

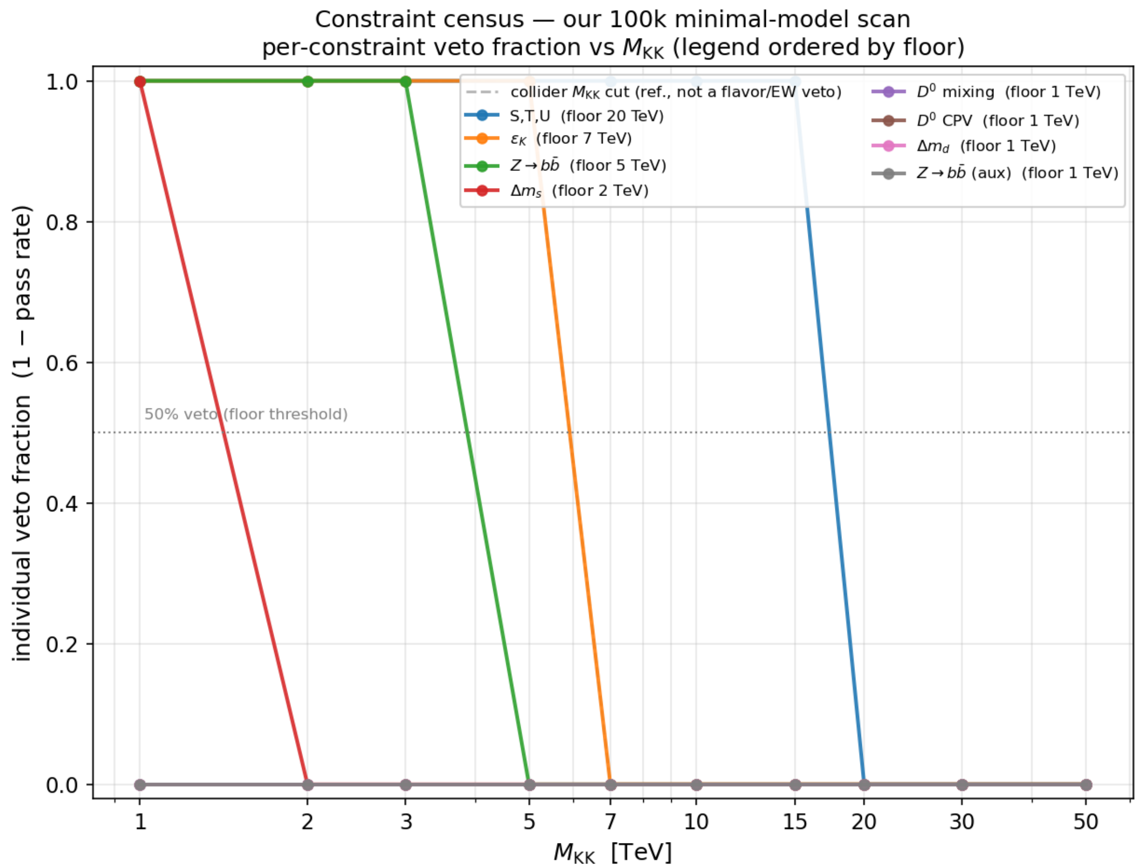
The split matters because the flavor constraints are tunable (you can align phases so the new-physics piece cancels, e.g. $\varepsilon_K \propto \mathrm{Im}\, M_{12} \to 0$), but $S, T, U$ is not: in minimal RS the oblique $T$ depends only on $M_{\mathrm{KK}}$ and the warp volume, with no Yukawa freedom. So even a fine-tuned minimal RS point cannot sit below about 18 to 20 TeV, and that wall is the oblique $S, T, U$ bound. This is exactly why custodial RS exists (it protects $T$).

| Constraint | Typical (median) floor | Existence (best/tuned) floor | Tunable? |
|---|---|---|---|
| $\varepsilon_K$ | ∼30 TeV (lane A anarchic; lane B ∼7, lane C ∼2) | below ∼1 TeV | yes (align I |
| $\Delta m_d$ ($B_d$) | few TeV | below ∼1 TeV | yes |
| $\Delta m_s$ ($B_s$) | few TeV | below ∼1 TeV | yes |
| $D^0$ | low | below ∼1 TeV | yes |
| $\Delta m_K$ | low | below ∼3 TeV | yes |
| $Z \to b\bar{b}$ (T010) | ∼5 TeV | ∼4.6 TeV | no (gauge, |
| $S, T, U$ (EW001) | ∼20 TeV | ∼18 to 20 TeV | no (no Yul |
| **Combined** | **∼30 TeV** ($\varepsilon_K$) | **∼18 to 20 TeV** ($S, T, U$) | |

**Existence vs typical, per constraint.** Min ratio-to-bound (best, fine-tuned point) and median ratio (typical point) vs $M_{\mathrm{KK}}$. The flavor curves crash well below 1 (tunable, no real floor); $S, T, U$ has min $\approx$ median and crosses 1 only near 18 to 20 TeV (irreducible).




**Our 100k scan, constraint census.** Per-constraint veto fraction vs $M_{\mathrm{KK}}$ from our minimal scan with all fixes in. Ranking: $S, T, U$ at 20, $\varepsilon_K$ at 7, $Z \to b\bar{b}$ at 5 (this was the old 25 to 30 TeV artifact before the fix), $\Delta m_s$ at 2. Dropping $Z \to b\bar{b}$ does not move the combined floor, so it is not the driver.

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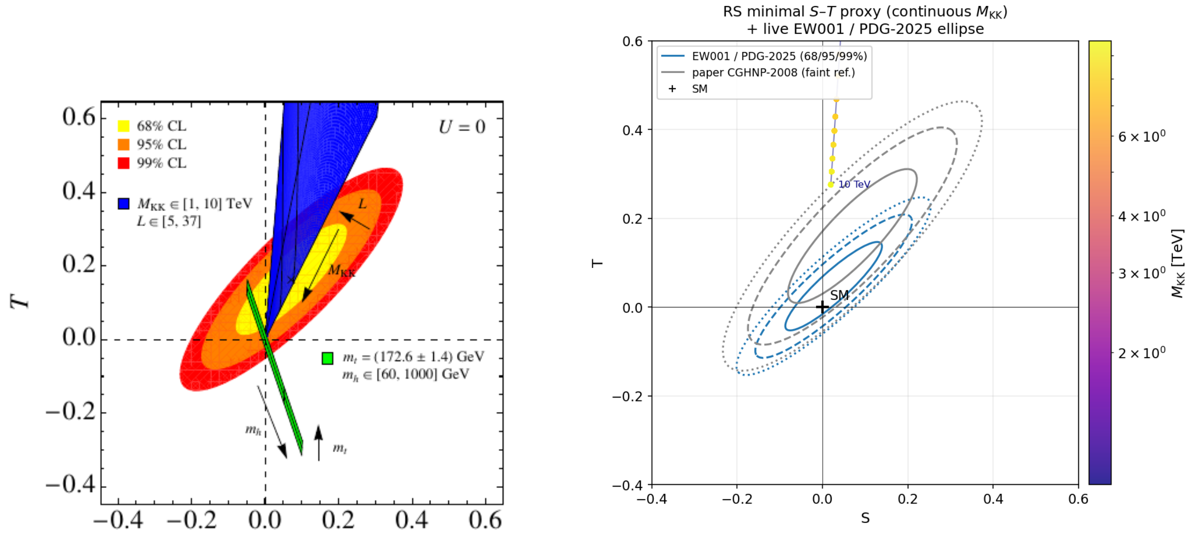
## 2. Reproductions (paper left, ours right)

Each comparison is the correctly cropped single paper panel (minimal RS / S1 anarchic) on the left, our reproduction on the right. Where the published figure was a multi-panel (a custodial companion, a Higgs-mass sweep, or the S2/S3/S4 scenarios), only the one panel we reproduce is shown.

### $S, T, U$ oblique (Casagrande et al. 0807.4937, Fig. 4)

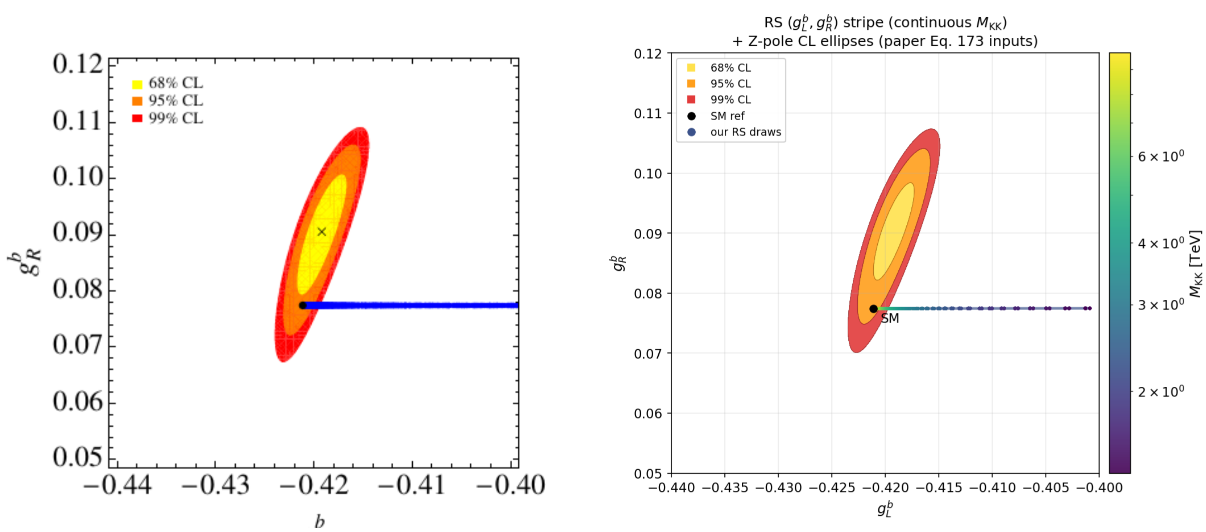
Our $S$-$T$ plane vs the paper. The RS minimal wedge climbs steeply in $T$ (the volume-enhanced $T$ problem) and leaves our PDG-2025 ellipse fast; grey is the 2008 ellipse for reference. This is the irreducible bound that sets the existence floor. The paper panel on the left is the minimal RS one (the blue wedge shooting up in $T$); the custodial companion panel, where the blue band stays pinned near $T = 0$, is not what we model and is dropped.




paper (left, 0807.4937 Fig. 4, minimal RS panel) vs ours (right)

## $Z \to b\bar{b}$ couplings (Casagrande et al. 0807.4937, Fig. 8)

Our $(g_L^b, g_R^b)$ plane vs the paper. The RS stripe pins at $g_R^b \approx 0.077$ and slides to less-negative $g_L^b$ as $M_{\rm KK}$ drops, staying right by the SM point. With the B1 fix this is gauge-dominated and weak (bites only to about 5 TeV), not the 25 to 30 TeV the bug produced. The paper panel on the left is the RS scatter/stripe; the companion $m_h = 60$ to 1000 GeV Higgs-mass sweep panel, which we do not compute, is dropped.




paper (left, 0807.4937 Fig. 8, minimal RS panel) vs ours (right)

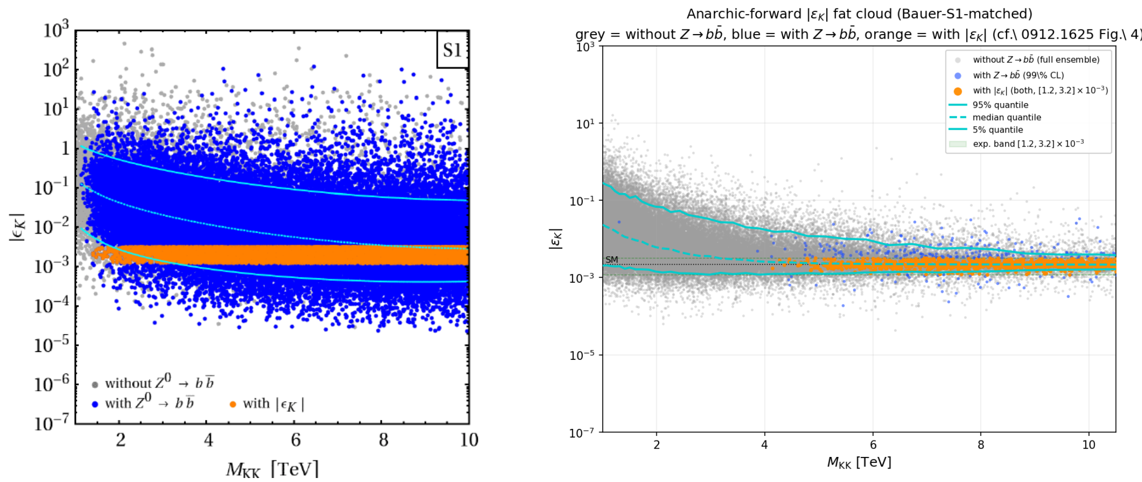
## $\varepsilon_K$, S1 standard (Bauer et al. 0912.1625, Fig. 4 top-left)

Our anarchic $|\varepsilon_K|$ cloud vs the paper, using Bauer's S1 draw setup ($Y_{\max} = 3$, anarchic Yukawas drawn uniform in modulus and phase, bulk $c$ scanned and fixed per draw by the Froggatt-Nielsen mass relations). One thing we had to get right: Bauer KEEPS only draws that reproduce the quark masses and CKM (his $\chi^2/\mathrm{dof} < 11.5/10$ cut, then no single observable off by more than $3\sigma$). Our forward scan computes every draw but flags whether it passes that mass+CKM gate, and the whole figure here, grey backdrop and the quantile curves and the blue/orange overlays, is now restricted to that passing subset. This matters: on the full unfiltered ensemble the median third-generation doublet $c_{Q_3}$ rails to the IR floor (about $+0.08$ in our convention) because the $\sim$89% of mass+CKM FAILING draws push $b_L$ very IR; once we apply the gate the median $c_{Q_3}$ rises to about $+0.28$, which sits comfortably inside Bauer's Table 1 value $c_{Q_3} = -0.34 \pm 0.32$ (their sign convention is opposite ours, so their $-0.34$ is our $+0.34$). To be honest about it, our gated median $+0.28$ is about 0.06 below Bauer's central $+0.34$, well within his $\pm 0.32$ one-sigma width; the distribution is broad and slightly tailed, exactly as Bauer notes for $c_{Q_3}$, so the central value is not sharply pinned. The point is that the gate moves the median the right way and by the right amount, from the IR floor up to the Bauer band. Pulling $c_{Q_3}$ more UV is also what weakens the $Z \to b\bar{b}$ shift, which is why the gated ensemble has a noticeably larger blue (Z-consistent) fraction at high $M_{\mathrm{KK}}$.

The cloud is fat, about 6 to 7 decades, and it is built as a HYBRID of two layers drawn from the SAME Bauer-S1 prior. The grey backdrop and the 5/50/95 percent quantile curves come from the FAT 990,000-draw `anarchic_bauer_S1` ensemble (112,024 mass+CKM-gated draws on the 45-tile $M_{\mathrm{KK}}$ grid from 1 to 20 TeV, clipped to 1 to 10.5 TeV here to match Bauer), which samples the $|\varepsilon_K|$ tails far better than the Z-companion run and so restores Bauer's headline per-tile spread (median about 3.7, max about 4.9 decades). The blue ($Z \to b\bar{b}$-consistent) and orange ($|\varepsilon_K|$-window) overlays come from the DENSE 202,500-draw `anarchic_bauer_s1_zbb` companion (22,880 mass+CKM-gated draws, of which 5,401 pass $Z \to b\bar{b}$ at 99% CL and about 5,000 also land in the $|\varepsilon_K|$ window). Both layers are the same anarchic Bauer-S1 prior, so the blue Z-passing points are a representative subset of the grey distribution; we no longer need the single-ensemble construction for blue density now that the companion carries 5,401 blue draws. The 5/50/95 percent quantile curves (cyan) are computed on that fat 990,000-draw ensemble; the 95 percent quantile enters the allowed band near 10 TeV, which is Bauer's headline number. Following Bauer's Fig. 4 caption the points are coloured by $Z \to b\bar{b}$ consistency: grey is the (mass+CKM-consistent) ensemble without the $Z \to b\bar{b}$ cut, blue is consistent with both $Z \to b\bar{b}$ couplings at 99% CL, and orange is consistent with both $Z \to b\bar{b}$ (99% CL) and $|\varepsilon_K| \in [1.2, 3.2] \times 10^{-3}$ (95% CL). The per-draw $Z \to b\bar{b}$ shift is the total mass-basis $[2, 2]$ entry (gauge-KK plus the Casagrande fermion-KK admixture) computed on the same anarchic draws by `scripts/anarchic_bauer_s1_zbb.py`. The paper figure is a 2x2 over scenarios S1 to S4; the left panel here is the S1 anarchic panel (top-left of the original).

One honest difference from Bauer, and it is a real result rather than a plotting artifact: in our panel the blue ($Z \to b\bar{b}$-consistent) points fill in only ABOVE about 4.5 TeV (the blue fraction is essentially zero below 4 TeV, crosses 1 percent near 4.5 TeV, and climbs from there), whereas Bauer's blue blankets the entire plane down to 1 TeV. The reason is that our corrected $Z \to b\bar{b}$ constraint is genuinely STRONGER than Bauer's 2009 version: we use the PDG-2025 $R_b/A_b$ inputs and the exact $Zb\bar{b}$ coupling shift, so low-$M_{\mathrm{KK}}$ draws that Bauer would have called $Z \to b\bar{b}$-consistent are vetoed in our evaluation. This is the same physics that pushes our $Z \to b\bar{b}$ floor to about 5 TeV (Section 1), so the blue onset near 4.5 TeV is exactly where it should be. We are not claiming a pixel-perfect match to Bauer's blue coverage; the cloud shape, decoupling, quantile floor and orange in-band stripe do match, but our blue is honestly more restrictive at low $M_{\mathrm{KK}}$.

This S1 standard scenario, anarchic Yukawas with INDEPENDENT per-generation bulk $c$'s fit to masses and CKM, is exactly what our production catalog scan implements (`quarkConstraints/fit.py` fits three independent down $c$'s; there is no common-$c$ alignment anywhere in the production path). So this panel is the apples-to-apples reproduction of our own model.

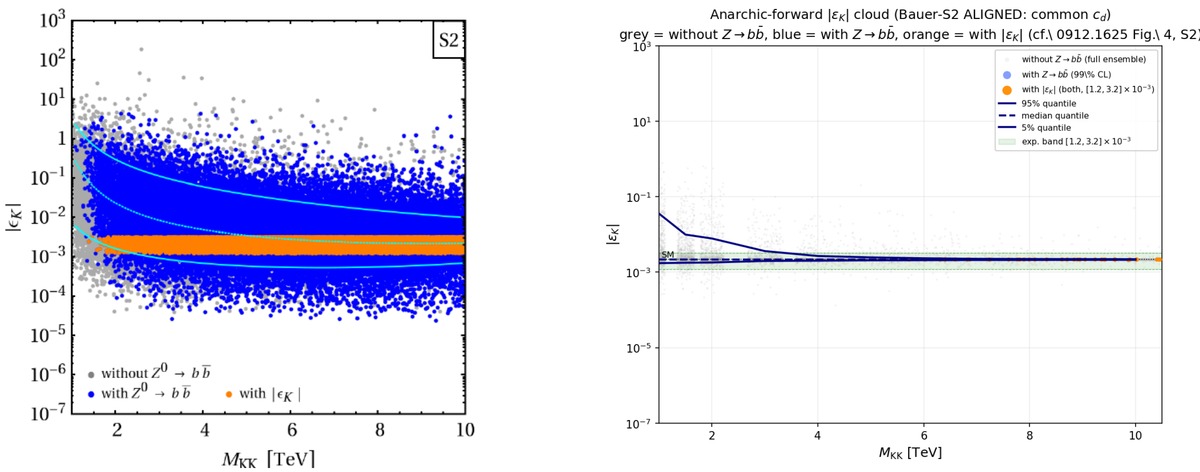



paper (left, 0912.1625 Fig. 4, S1 panel) vs ours (right)

## $\varepsilon_K$, S2 aligned (Bauer et al. 0912.1625, Fig. 4 top-right)

For completeness, here is Bauer's OTHER scenario, S2 "aligned". It is the same anarchic setup ($Y_{\max} = 3$, same mass+CKM gate) with one extra assumption: a COMMON right-handed down-type bulk mass $c_{d_1} = c_{d_2} = c_{d_3}$, imposed by a $U(3)$ flavour symmetry on the RH down quarks. That alignment suppresses the dangerous $\Delta F = 2$ down-sector mixing, so the $\varepsilon_K$ cloud collapses toward the experimental band and the orange (in-band) fraction blows up relative to S1. You can see it in both panels: the orange stripe in S2 is far fatter than in S1, and the median $\varepsilon_K$ sits much closer to the band. Our S2 panel is built byte-identically to the S1 one (same draws, same gate, same $Z \to b\bar{b}$ colouring) except the scan averages the three down $c$'s to a single value per draw (`scripts/anarchic_bauer_s1.py`, `common_cd=True`). The paper panel on the left is the top-right (S2) of the original 2x2.

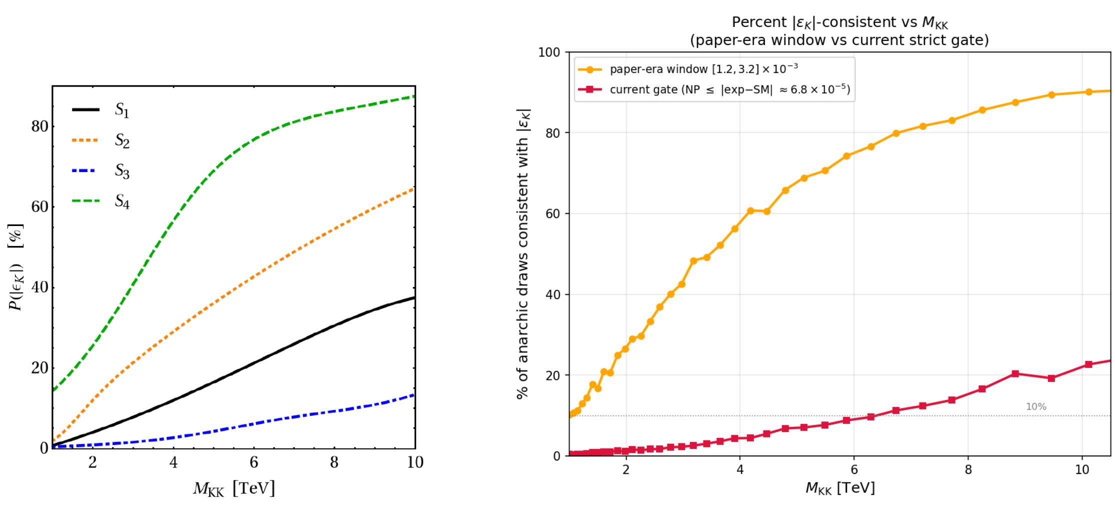
Honest note on what this is: S2 is a DIFFERENT model from the one we actually scan. Our production catalog uses independent down $c$'s (it is S1 standard, confirmed above), so it does NOT get the S2 alignment suppression for free; the $\varepsilon_K$ floor our scan reports is the S1 one. S2 is shown here only to reproduce Bauer's full figure and to make the point explicit: the way to soften the $\varepsilon_K$ tension in this class of models is an extra down-sector flavour symmetry, which is a model-building choice (a different model) rather than a free consequence of warped geometry. We are not claiming our scan sits in S2.




paper (left, 0912.1625 Fig. 4, S2 aligned panel) vs ours (right)

## Consistency fraction (Bauer et al. 0912.1625, Fig. 5)

Fraction of anarchic points consistent with $|\varepsilon_K|$ vs $M_{\rm KK}$, paper-era window vs current strict gate. The rising $1/M_{\rm KK}^2$ decoupling is the same shape as the paper. The paper figure has three panels, $P(|\varepsilon_K|)$, $P(Z \to b\bar{b})$, and $P(\text{total})$; the left panel here is the $P(|\varepsilon_K|)$ one (top-left of the original), since that is the quantity we plot. We reproduce the S1 (black solid) curve in it; the $Z \to b\bar{b}$ and total panels are dropped.

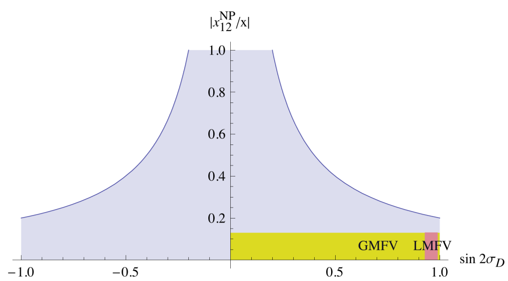


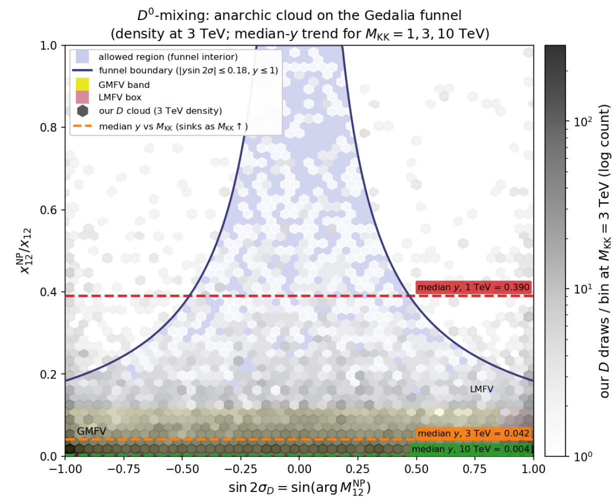

paper (left, 0912.1625 Fig. 5, $P(|\varepsilon_K|)$ panel) vs ours (right)

## $D^0$ mixing (Gedalia, Grossman, Nir, Perez 0906.1879, Fig. 1)

Our anarchic $D$-meson cloud on the Gedalia funnel vs the paper, cleaned up to a single 3 TeV density plus the median-$y$ trend at 1, 3, 10 TeV. The cloud sinks into the allowed funnel as $M_{\rm KK}$

rises. The paper figure is a single panel (the allowed funnel with the GMFV and LMFV bands), kept as is. (Optional figure: say the word if you would rather drop it.)

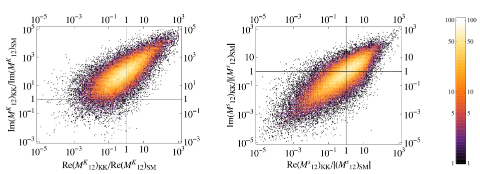


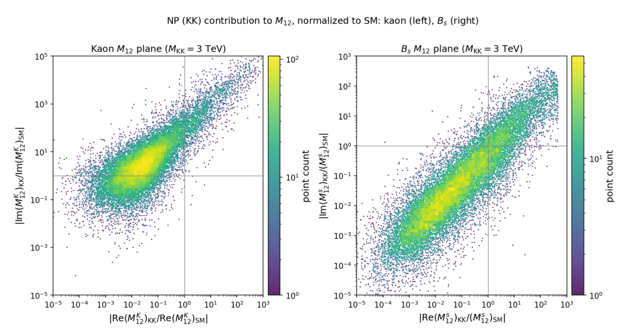




paper (left, 0906.1879 Fig. 1, single funnel panel) vs ours (right)

## $\Delta F = 2$ amplitudes (Blanke et al. 0809.1073, Fig. 2)

Re vs Im of the KK contribution to $M_{12}$, for $K$ (left sub-panel) and $B_s$ (right sub-panel). For kaons the imaginary part dominates the real part by about 100× (the $\varepsilon_K$ problem); ours is 106×. The paper Fig. 2 is exactly these two systems, $K$ and $B_s$ (it has no $D$ or $B_d$ sub-panels), so both are kept on the left.






paper (left, 0809.1073 Fig. 2, $K$ and $B_s$ panels) vs ours (right)

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**Dropped on purpose:** the $B_d$ and $B_s$ phase planes (the old Figs 6 and 7). Our model clouds are fine, but the experimental CL contours we drew there are only Gaussian approximations of the CKMfitter regions, not the exact published contours, so they are not a clean reproduction. Left out rather than shown as if exact.